



3 Ton Melting with CaO Desulfurization of Ni-Base Single Crystal Superalloy TMS-1700, Simulating a Recycling of Used Turbine Blades

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Abstract

In order to investigate the applicability of a direct and complete recycling method to commercial-scale ingot production, 3 tons of a Ni-base single crystal (SC) superalloy, TMS-1700(MGA1700) was melted and desulfurized by CaO during the melting. Sulfur content in the molten alloy was reduced from about 23 ppm to about 2 ppm within 60 min after adding granular CaO to the molten alloy. Microstructural observations using SEM and EPMA showed no presence of inclusions caused by CaO addition. Creep rupture lives of the recycle-simulated TMS-1700 were equivalent to that of the standard TMS-1700 under the conditions from 800 °C/735 MPa to 1100 °C/137 MPa. The recycle-simulated TMS-1700 exhibits even better oxidation resistance compared with the standard TMS-1700, which has better oxidation resistance than CMSX-4TM. Thus, it became clear that the CaO desulfurization improves the oxidation resistance of Ni-base superalloys. High cycle fatigue (HCF) properties of the standard and the recycle-simulated TMS-1700 were equivalent. From the results described above, it has been suggested that the application of direct and complete recycling method to commercial-scale ingot production is feasible.

Keywords

Ni-base single crystal superalloys • Recycle • CaO desulfurization • Creep • Oxidation resistance

Introduction

Ni-base single crystal (SC) superalloys are widely used as materials for high temperature components in many aero-engines and land-based gas turbines. To improve the thermal efficiency of gas turbines, the inlet gas temperatures of gas turbines are increased. To achieve this, materials with higher temperature capabilities are needed for turbine blades and vanes. Kawagishi et al. developed the advanced Ni-base SC superalloys [1–3]; however, higher material cost of advanced Ni-base SC superalloys due to precious alloying elements, such as Re and Ru, has limited their wider applications.

The authors are working on the establishment of a direct and complete recycling method that enables a drastic reduction in lifetime costs of Ni-base SC superalloys at the project of Japan Science and Technology Agency (JST), under “Advanced Low Carbon Technology Research and Development Program (ALCA), JPMJAL1302” [4]. We have succeeded in direct and complete recycling trial of scrapped PWA1484 turbine blades using CaO crucible melting up to 20 kg level [5]. In the research, S contamination in scrapped PWA1484 turbine blade, typical concentration being about 10 ppm, was found to be detrimental especially in oxidation resistance [6, 7]. The S contamination was successfully refined in a CaO crucible to the original ingot level, 2 ppm. It was also demonstrated that the influence of bond coating material can be well canceled out by a composition adjustment which we call counter alloying. Other studies have been reported regarding the desulfurization of Ni-base superalloys. Degawa et al. revealed that CaO has the best stability and the best efficiency for desulfurization among all crucibles tested such as Al₂O₃, MgO, ZnO, and CaO [8]. Sidorov et al. have reported that addition of Ca or rare earth elements during melting in a CaO crucible reduces the S content in molten Ni-base superalloys [9, 10]. However, these results were limited to laboratory-scale tests, and commercial-scale capability has not been demonstrated.

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In this study, in order to investigate the applicability of the direct and complete recycling method to the commercial-scale ingot production, 3 tons of SC superalloy, TMS-1700 (MGA1700) [11], was melted and desulfurized by CaO during the melting. This test-recycling was performed using a commercial melting furnace, which has been operating for commercial-ingot production of Ni-base superalloys. TMS-1700 is a Re-free SC superalloy developed in collaborative work with Mitsubishi Heavy Industries, to realize a 1700 °C-class land-based gas turbine. The feasibility of the desulfurized superalloy was investigated by comparing mechanical and environmental properties of it with the standard and the recycle-simulated ones of TMS-1700.

Three Tons Melting with CaO Desulfurization

The nominal composition of the TMS-1700(MGA1700) is shown in Table 1. Using a commercial-scale low-frequency induction furnace (150 Hz, 1500 kW) with a maximum capacity of 6 tons, 3 tons of TMS-1700 was melted using raw materials. A MgO crucible was used in this study. FeS was intentionally added so that the concentration of S was about 25 ppm in order to simulate heavily contaminated scrapped materials; this S content was more than 3 times as high as the actual analysis of scrapped PWA1484 turbine blades, 10 ppm [5]. Sixteen minutes after FeS addition, 15 kg of CaO granules having the average particle size of about 3 mm were added without breaking the vacuum to start the desulfurization.

Figure 1 shows the S concentration in the alloy during melting, desulfurization, and ingot casting processes.

Greek numbers in the figure indicate the points after the CaO addition when chemical analyses were performed. The concentration of S drastically decreased from 23 ppm to about 3 ppm 30 min after CaO granules were added, and it was stabilized at 2 ppm at 60 min and later. The analytical compositions at the point I, II, and III were also shown in the lower part of Table 1. The inductive plasma emission spectrometry (Agilent technologies: Agilent720ES) was applied for alloying elements, and the glow discharge mass spectrometry (LECO: CS444LS) was applied for S. The

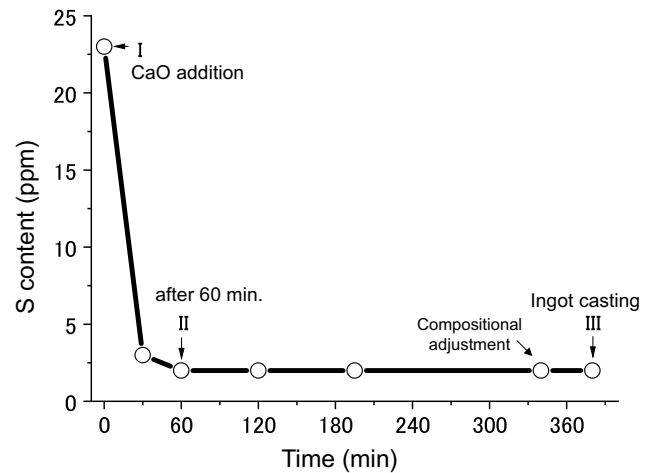


Fig. 1 Sulfur concentration of molten metal during the CaO desulfurization process. Greek numbers and O marks indicate the timing of the chemical analysis

content of Fe in the standard and CaO desulfurized TMS-1700 were also analyzed. The concentrations of them were 0.01 and 0.02 wt%, respectively. These concentrations were within the alloy specification, 0.1 wt%, that effect on mechanical and environmental properties described below might be negligible. The compositional adjustment of the molten metal was performed before the ingot casting, because the compositional change of the molten metal occurs during the CaO desulfurization process since the decrement of Al due to the formation of Al_2O_3 [12] and also due to evaporation of Cr. To determine the adjustment range of alloying elements, the creep strength of the alloy before the adjustment was predicted using our alloy design program [13]. Table 2 shows the prediction results of the creep rupture lives and microstructural parameters using the alloy design program for the compositions at point I, II, and III in Fig. 1. Using these results, we ensured that the creep strength was within the target range.

After the ingot casting, no CaO residues were observed in the tundish. In addition, the cross-sectional analysis of crucible wall after ingot casting has conducted using the electron probe micro-analyzer (Shimadzu, EPMA-1610).

Figure 2a shows a part of crucible wall after ingot casting and (b) shows the cross-sectional image of the crucible wall.

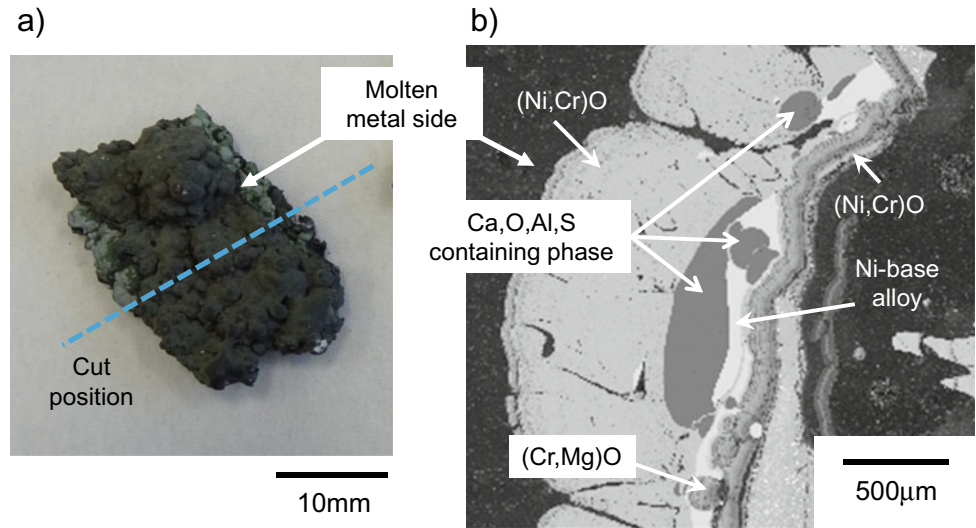
Table 1 Nominal composition of TMS-1700, compositions of molten metal at point I, II, and III are also shown (wt% except S, Ni; balanced)

	Compositions (wt%, Ni; balanced)								
	Co	Cr	Mo	W	Al	Ta	Hf	Si	S (ppm)
Nominal	0	9.00	0.60	7.60	5.40	10.00	0.10	0.04	–
I. 60 min. after FeS addition	0	8.98	0.59	7.58	5.30	10.18	0.07	0.04	23
II. 60 min. after CaO addition	0	8.40	0.60	7.69	5.21	10.33	0.09	0.04	2
III. Ingot	0	9.17	0.60	7.52	5.41	9.93	0.08	0.04	2

Table 2 Predicted creep rupture lives and structural parameters for the compositions at point I, II, and III shown in Fig. 1

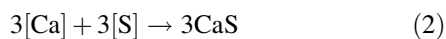
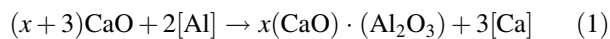
	Creep rupture (h)		V_f of γ' phase (%)		Window (°C)
	900 °C/392 MPa	1000 °C/245 MPa	900 °C	1000 °C	
I. 60 min. after FeS addition	205.9	73.7	65	61	18.6
II. 60 min. after CaO addition	189.8	67.4	63	60	18.3
III. Ingot	208.3	76.6	66	63	19.2

Fig. 2 **a** A part of crucible wall after the ingot casting, **b** cross sectional image of crucible wall



From the cross-sectional analysis, there was no CaO, but a (Ca, O, Al, S) containing phase, (Cr, Mg) oxides were covered by thick (Ni, Cr) oxides. The analysis result revealed that (Ca, O, Al, S) composites, (Cr, Mg) oxides were deposited on the crucible wall initially, and thick (Ni, Cr) oxides were formed during the crucible wall cooling. This result indicates that desulfurization reaction was almost finished after 60 min, and it suggests that S in the melt reacts with CaO to form CaS and [O], and then, Al in the melt reacts with [O] and CaO to generate calcium aluminate on the crucible surface [8, 14–16].

The desulfurization reaction of Ni-base superalloy melt using solid CaO can be expressed as following formulas.



Hereafter, the CaO desulfurized TMS-1700 is referred to as the recycle-simulated TMS-1700.

Mechanical and Environmental Properties

Mechanical and environmental properties of the recycle-simulated TMS-1700 were investigated by the creep rupture tests, the cyclic oxidation tests, the high cycle fatigue tests,

and the fracture toughness tests. The results were evaluated by comparison with the results for the standard TMS-1700.

By using the standard and the recycle-simulated TMS-1700 ingots, SC bars were cast by the vacuum-induction melting and standard directional solidification casting. Crystal orientations of SC bars were measured by the X-ray Laue method. Then, the samples within eight degrees from [001] orientation were solution heat treated and aged. No residual eutectics and no visible TCP phases were observed in their microstructures.

Test specimens were machined from the heat-treated samples. Figure 3 shows the test specimen configurations for (a) creep rupture, (b) cyclic oxidation, and (c) high cycle fatigue.

Creep Rupture Test

The creep tests for the standard and the recycle-simulated TMS-1700 were conducted under four conditions of 800 °C/735 MPa, 900 °C/392 MPa, 1000 °C/245 MPa, and 1100 °C/137 MPa. Figure 4a–d shows the creep curves of the standard and the recycle-simulated TMS-1700. Although a slight difference was observed at 1100 °C/137 MPa, it became clear that the creep rupture life of each alloy was almost the same under all testing conditions in this study.

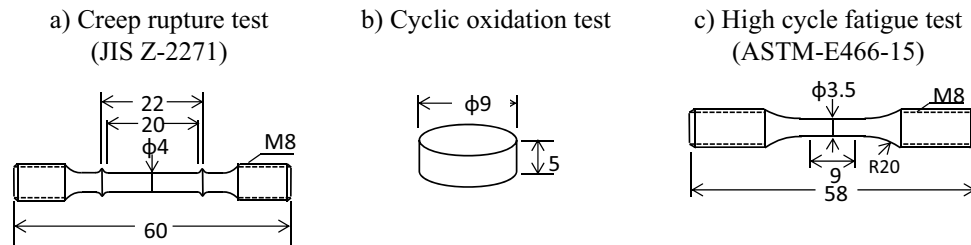


Fig. 3 Shapes of specimens for **a** creep rupture test, **b** cyclic oxidation test, and **c** high cycle fatigue test (unit; mm)

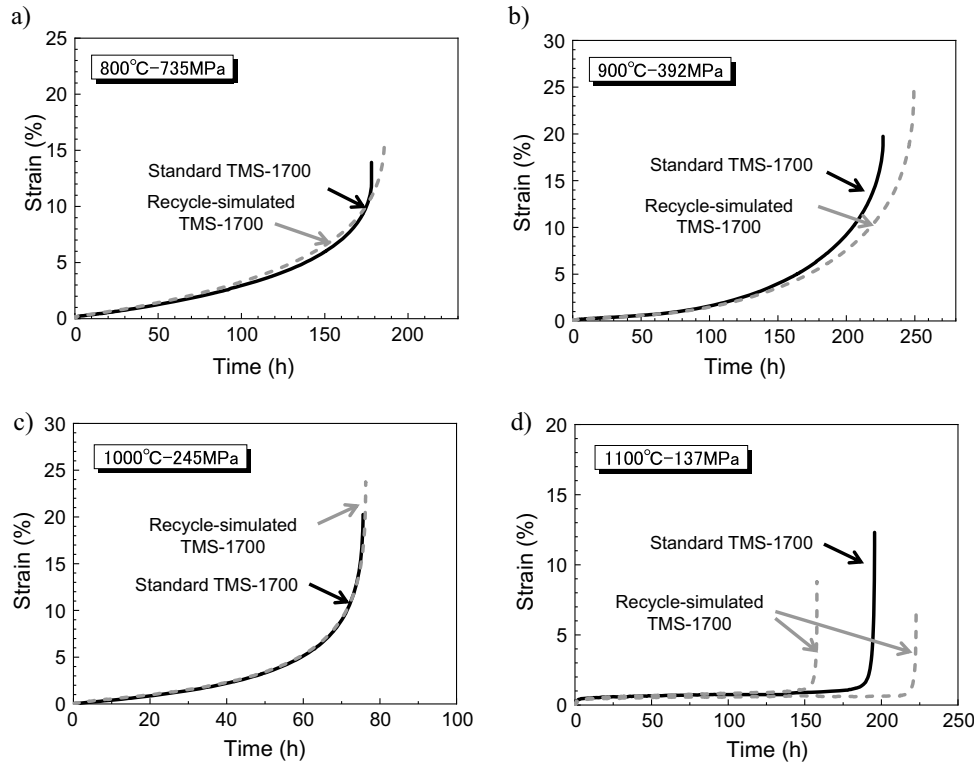


Fig. 4 Creep curves of the standard and the recycle-simulated TMS-1700, **a** at 800 °C/735 MPa, **b** at 900 °C/392 MPa, **c** at 1000 °C/245 MPa, and **d** at 1100 °C/137 MPa

Figure 5 shows the Larson–Miller plot ($C = 20$) of the standard and the recycle-simulated TMS-1700. Regarding the Larson–Miller plot, there is no significant difference between these two alloys. The creep strength of the recycle-simulated TMS-1700 is similar to the standard one. In addition, the long-term creep tests of the standard and recycle-simulated TMS-1700 were conducted for use in land-based turbines. As shown in Fig. 6, it was found that the standard and the recycle-simulated TMS-1700 had the same creep strength even under the long-term creep conditions of 900 °C/245 MPa and 850 °C/294 MPa.

Cyclic Oxidation Test

The oxidation resistance of the recycle-simulated TMS-1700 was evaluated by the cyclic oxidation test at 1100 °C/1 h heating and 1 h air cooling per cycle. Figure 7 shows the result of cyclic oxidation test of each alloy with the oxidation behavior of a commercial alloy, CMSX-4TM (ULS) [17] as a reference. The standard TMS-1700 has good oxidation resistance further better than CMSX-4TM. It should be noted that the recycle-simulated TMS-1700 showed better oxidation

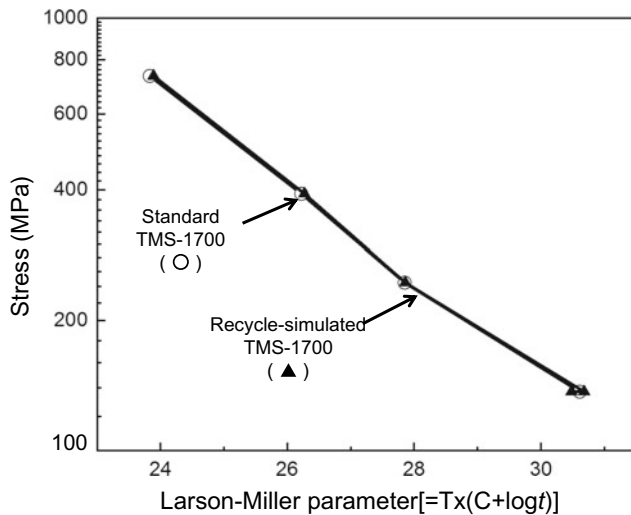


Fig. 5 Larson-Miller plot ($C = 20$) of the standard and the recycle-simulated TMS-1700

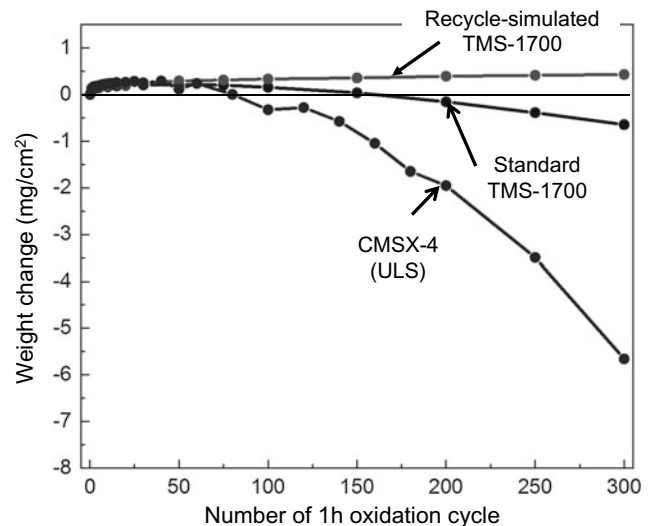


Fig. 7 Results of cyclic oxidation tests of the standard and the recycle-simulated TMS-1700. The oxidation behavior of commercial alloy, CMSX-4TM(ULS), is also plotted

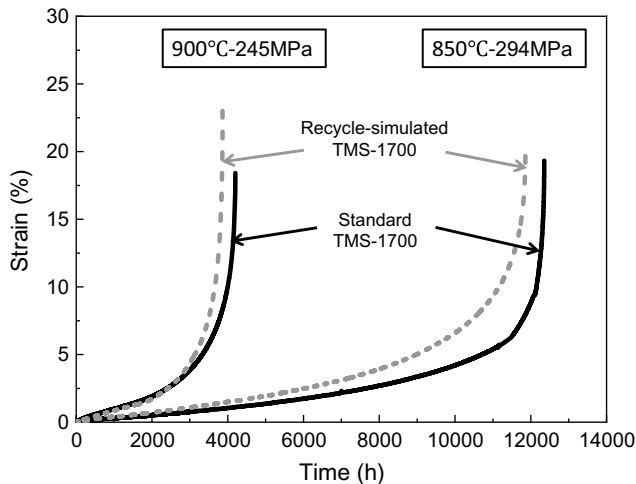


Fig. 6 Creep curves of the long-term creep tests of the standard and the recycle-simulated TMS-1700 at 850 °C/294 MPa, and at 900 °C/245 MPa

resistance than the standard TMS-1700 even though the S concentration of both alloys is similar to each other; 2 ppm. These results revealed that the CaO desulfurization process shows the additional effect, improvement of oxidation resistance. The mechanism of this oxidation resistance improvement is becoming clear in our separate works [18, 19].

High Cyclic Fatigue (HCF) Test

Using two specimens of the standard and six specimens of the recycle-simulated TMS-1700, HCF tests at 1000 °C and maximum stress of 300 MPa, $R = 0$ were conducted.

Figure 8 shows the HCF test results for each alloy. For reference, CMSX-4TM results under the different test conditions are also plotted. The number of cycles to failure of both the standard and the recycle-simulated TMS-1700 was similar. Fracture surface observations of both alloys are shown in Fig. 9a–d. Cracks initiated and propagated from cast voids, and no inclusion-initiated cracks were observed in both alloys. From these results, it was concluded that HCF properties of recycle-simulated alloy were similar to standard one and inclusions possibly introduced by CaO granules addition did not affect the HCF property of the recycle-simulated TMS-1700.

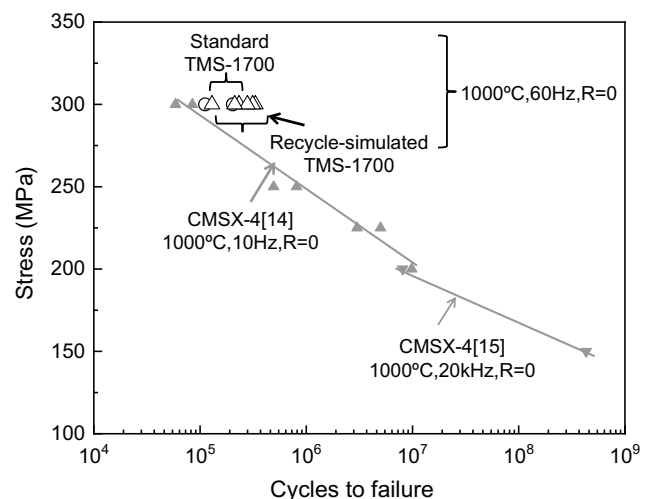


Fig. 8 Results of HCF tests of the standard and the recycle-simulated TMS-1700 under 1000 °C, maximum stress of 300 MPa, $R = 0$

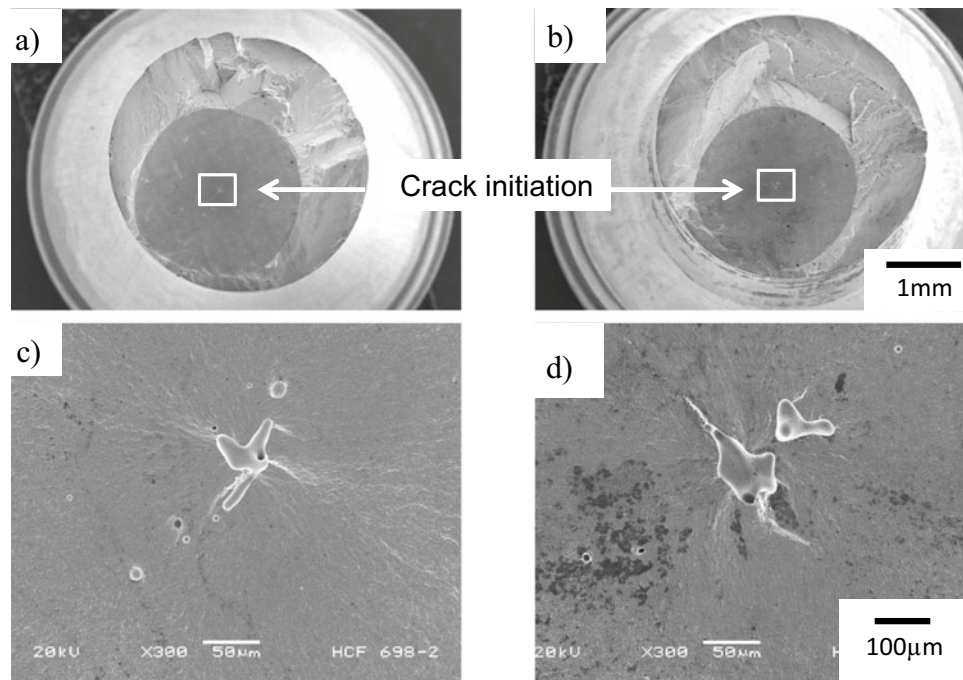


Fig. 9 Microstructures of fractured surfaces after HCF tests. **a** and **c** the standard TMS-1700 and **b** and **d** the recycle-simulated TMS-1700. **c** and **d** are the close-up images of the initiation sites

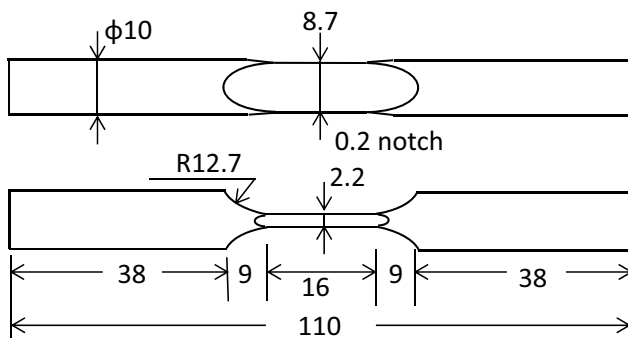


Fig. 10 Shape of SENBH specimen with 0.2 mm notch for the fatigue crack propagation test (ASTM E647-15 base)

Fracture Toughness Tests

The fracture toughness value of Ni-base superalloys is one of the most important mechanical properties along with strength and ductility. The fracture toughness of the standard and the recycle-simulated TMS-1700 were measured and compared to each other [20].

The fracture toughness values of alloys were derived as ΔK_{IC} from the crack propagation rate and the stress intensity factor obtained from the fatigue crack propagation test. Figure 10 shows the geometry of SENBH specimen with 0.2 mm notch. The fatigue crack propagation tests based on ASTM E647-15 were carried out at room temperature,

frequency of 5 Hz, stress ratio of $R = 0.1$, and constant load condition. The fracture toughness values of the standard TMS-1700 were in the range from 55 to 70 $\text{MPa m}^{1/2}$ and that of the recycle-simulated TMS-1700 was from 55 to 85 $\text{MPa m}^{1/2}$; the values were equivalent to each other. As shown in Fig. 11, the fracture toughness values of both alloys are in the same range as other Ni-base superalloys.

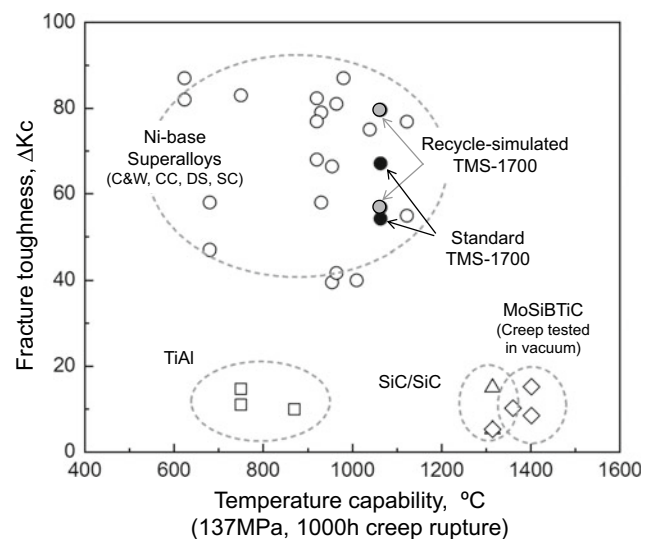


Fig. 11 Fracture toughness of the standard and the recycle-simulated TMS-1700. Fracture toughness of commercial and experimental Ni-base superalloys, TiAl, SiC/SiC, and MoSiBTiC are also plotted in this figure

Fracture toughness values of TiAl, SiC/SiC, and MoSiB/TiC are also shown in the same figure just for references, the values being less than 20 MPa m^{1/2} [21–24].

Conclusions

In order to investigate the applicability of a direct recycling method to commercial-scale ingot production, simulated recycle, 3 tons melting of FeS added TMS-1700 with CaO desulfurization followed by ingot casting, was performed. Mechanical and oxidation properties of the recycle-simulated TMS-1700 cast from the resultant ingot were evaluated. The results obtained were as follows:

1. By adding CaO granules into 3 tons molten TMS-1700 alloy containing 23 ppm S, desulfurization was achieved; S content in the molten TMS-170 was reduced to about 2 ppm within 60 min after CaO addition.
2. Recycle-simulated TMS-1700 showed equivalent creep strength to that of standard TMS-1700 both in short- and long-term creep test conditions.
3. Recycle-simulated TMS-1700 exhibited excellent oxidation resistance, slightly better than standard TMS-1700. Both of them showed better oxidation resistance than a standard commercial alloy, CMSX-4TM. It was revealed that the CaO desulfurization shows additional effect, improvement in the oxidation resistance.
4. HCF properties of the standard and recycle-simulated TMS-1700 were equivalent. No HCF degradation was observed originated from possible inclusions caused by CaO addition.
5. Fracture toughness values of the standard and recycle-simulated TMS-1700 were equivalent to each other, the values being in the range from 55 to 85 MPa m^{1/2}, which were in the same range as other Ni-base superalloys.

As described above, it has been suggested that the application of the direct and complete recycling method to large-scale ingot production is feasible. This method has a potential not only for cost reduction but also for securing the resource issues of advanced Ni-base SC superalloys, especially for alloys containing precious alloying elements such as Re and Ru. Consequently, it is expected that the direct and complete recycling method promote wider applications of such advanced Ni-base SC superalloys in aeroengines and land-based gas turbines for increasing their inlet gas temperatures aiming at the higher thermal efficiencies.

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